Lecture 20 Scribe

L20.pdf
PDF

Write a detailed lecture scribe for exam preparation based entirely on the given material. Include every concept, definition, and assumption, and clearly explain the reasoning behind each along with their logical connections. Ensure that all steps and arguments are presented in a structured and easy-to-follow manner. Do not add any external information, examples, or interpretations beyond the provided content. The final output should be clear, well-organized, and comprehensive, allowing a student to revise effectively and reconstruct all concepts and solutions without needing additional resources.

Here is your **detailed lecture scribe for Lecture 20: Linear Transformations and Expectations of Multiple Random Variables**, constructed **strictly from the provided PDF** . Everything is organized step-by-step so you can **revise and reconstruct all concepts for exams**.

📘 Lecture 20: Linear Transformations and Expectations of Multiple RVs

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1. Random Vector Representation

Definition

For multiple random variables $X_1, X_2, \ldots, X_N$, they are represented in **vector form**:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Key Points

- This vector is called a **random vector**.
- It provides a **compact representation of multiple random variables**.
  📌 (From page 3)

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2. Mean Vector

Definition

The **mean vector** of a random vector X is:

$$\mu_X = E[X]$$

Interpretation

- Represents the **average value of the random vector**.
- Each element corresponds to the mean of one random variable.

Structure

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

📌 (From pages 4–5)

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3. Correlation Matrix

Definition

The **correlation matrix** is defined as:

$$R_{XX} = E[XX^T]$$

Element-wise form

$$R_{XX}(i,j) = E[X_i X_j]$$

Matrix Form

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots \\ E[X_2X_1] & E[X_2X_2] & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Property

- R_{XX} is a **symmetric matrix**.

📌 (From pages 6–7)

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4. Covariance Matrix

Definition

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

Expanded Form

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Key Property

- **Symmetry**:

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

📌 (From pages 8–9)

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5. Linear Transformation of Random Vectors

Model

$$Y = AX + b$$

Components

- A: Transformation matrix (scales/rotates)
- b: Constant vector (shifts)
- X: Input random vector

Expanded Form

Each element of Y is:

$$Y_i = \sum a_{ij} X_j + b_i$$

📌 (From page 10)

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6. Mean of Transformed Vector

Definition

$$\mu_Y = E[Y]$$

Derivation Steps

1. Substitute:

$$Y = AX + b$$

2. Take expectation:

$$\mu_Y = E[AX + b]$$

3. Apply linearity:

$$E[AX] + E[b]$$

Final Result

$$\mu_Y = A\mu_X + b$$

📌 (From pages 11–13)

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7. Correlation Matrix Transformation

Definition

$$R_{YY} = E[YY^T] = E[(AX + b)(AX + b)^T]$$

Expansion

$$R_{YY} = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

Applying Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^T A^T + bb^T$$

Final Result

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

📌 (From pages 14–17)

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8. Covariance Matrix Transformation

Key Observation

$$Y - \mu_Y = A(X - \mu_X)$$

Derivation

$$C_{YY} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

Substitute:

$$C_{YY} = E[A(X - \mu_X)(X - \mu_X)^T A^T]$$

Final Result

$$C_{YY} = AC_{XX}A^T$$

Important Note

- **Shift b does NOT affect covariance**

📌 (From pages 18–20)

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9. Summary of Linear Transformation

If:

$$Y = AX + b$$

Then:

Property	Result
Mean	$\mu_Y = A\mu_X + b$

Property	Result
Covariance	$C_{YY} = AC_{XX}A^T$
Correlation	$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$

📌 (From page 21)

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10. Numerical Example

Given (page 22)

- $\mu_X = [2, -1, 1]^T$
- Matrix A
- Correlation matrix R_{XX}

(a) Mean Vector

Using:

$$\mu_Y = A\mu_X$$

Result

$$\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

📌 (Pages 23–24)

(b) Correlation Matrix

$$R_{YY} = AR_{XX}A^T$$

Result

$$R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

📌 (Pages 25–28)

(c) Covariance Matrix

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

Result

$$C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

📌 (Pages 29–32)

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11. Gaussian Distributions

(a) Univariate Gaussian (1D)

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

📌 (Page 34)

(b) Joint Gaussian (2D)

- Defined for two variables using mean and correlation.

📌 (Page 35)

(c) Multivariate Gaussian (N-D)

For vector X:

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

Components

- μ_X: Mean vector
- C_{XX}: Covariance matrix

📌 (Pages 36–37)

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12. Special Case: Uncorrelated Gaussian Variables

Key Condition

$$\mathrm{Cov}(X_i, X_j) = 0 \quad \forall i \neq j$$

Result

- Covariance matrix becomes **diagonal**
- Joint PDF becomes:

$$f_X(x) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

Interpretation

- Joint distribution = product of individual Gaussians

📌 (Pages 48–51)

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13. Homework Problem

Statement

Prove:

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$$

📌 (Page 52)

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14. Case Study: Smart City System

Setup

- System predicts traffic using multiple variables.
- Define vector:

$$X = [X_1, X_2, X_3, X_4, X_5]^T$$

Variables (page 55)

- X_1: Traffic flow
- X_2: Speed
- X_3: Rainfall

- X_4: Temperature
- X_5: Pollution

Mean Vector

$$\mu = E[X]$$

Represents **average system conditions**.
📌 (Page 56)

Covariance Matrix

- Represents **relationships between variables**
- Called "**dependency map of the city**"

Observations (page 57)

- Rain ↑ → Traffic ↑
- Rain ↑ → Speed ↓
- Traffic ↑ → Pollution ↑
- Temperature ↑ → Pollution ↓

Joint Gaussian Assumption

$$X \sim \mathcal{N}(\mu, \Sigma)$$

Meaning

System is defined by:

- Mean → typical conditions
- Covariance → interdependencies

📌 (Page 58)

Linear Transformation (Congestion Index)

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

Results

- Y is also Gaussian
- Mean:

$$E[Y] = a^T \mu$$

- Variance:

$$\mathrm{Var}(Y) = a^T \Sigma a$$

📌 (Page 59)

Final Insight (page 61)

If rainfall increases:

- Speed ↓
- Traffic ↑
- Pollution ↑

✅ FINAL REVISION SUMMARY

MUST REMEMBER (Exam Focus)

1. Random vector representation
2. Mean vector: $\mu_X = E[X]$
3. Correlation: $R_{XX} = E[XX^T]$
4. Covariance: $C_{XX} = E[(X-\mu)(X-\mu)^T]$
5. Linear transformation: $Y = AX + b$
6. Mean transform: $\mu_Y = A\mu_X + b$
7. Covariance transform: $C_{YY} = AC_{XX}A^T$
8. Correlation transform formula
9. Multivariate Gaussian PDF
10. Diagonal covariance ⇒ independence-like behavior

If you want, I can:

✅ Convert this into **exam-ready short notes (1-page revision)**

✅ Solve the **homework proof step-by-step**

✅ Give **important expected questions for exam**

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Here is your **complete, structured lecture scribe (strictly based on the provided PDF)** .
It includes **all concepts, definitions, derivations, assumptions, and logical connections** exactly as presented, organized for **exam preparation and full reconstruction**.

📘 Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

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1. Statistical Representation of Random Vectors

1.1 Vector Form of Multiple Random Variables

For random variables $X_1, X_2, \ldots, X_N$, define:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Definition

- X is called a **random vector**.

Purpose

- Provides a **compact representation of multiple random variables**.

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2. Mean Vector

2.1 Definition

$$\mu_X = E[X]$$

Interpretation

- Represents the **average value of the random vector**.

Structure

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

Key Understanding

- Each element corresponds to the mean of one random variable.
- Thus, expectation extends **component-wise** to vectors.

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3. Correlation Matrix

3.1 Definition

$$R_{XX} = E[XX^T]$$

3.2 Element-wise Form

$$R_{XX}(i,j) = E[X_i X_j]$$

3.3 Matrix Representation

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \cdots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \cdots & E[X_NX_N] \end{bmatrix}$$

Property

- R_{XX} is a **symmetric matrix**.

Reasoning

- Since $E[X_iX_j] = E[X_jX_i]$, symmetry follows directly.

---

4. Covariance Matrix

4.1 Definition

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

4.2 Expanded Structure

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

4.3 Property

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

Reasoning

- Covariance definition is symmetric in its arguments.

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5. Linear Transformation of Random Vectors

5.1 Model Definition

$$Y = AX + b$$

Components

- A: Transformation matrix
- b: Constant vector
- X: Input random vector

Interpretation

- A: scales and rotates the data
- b: shifts/translates the center

5.2 Element-wise Representation

For N-dimensional vectors:

$$Y_i = a_{i1}X_1 + a_{i2}X_2 + \cdots + a_{iN}X_N + b_i$$

Logical Connection

- Each output component is a **linear combination** of input variables plus a shift.

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6. Mean of Transformed Vector

6.1 Definition

$$\mu_Y = E[Y]$$

6.2 Derivation

Step 1: Substitute transformation

$$\mu_Y = E[AX + b]$$

Step 2: Apply linearity of expectation

$$\mu_Y = E[AX] + E[b]$$

Step 3: Simplify

- $E[AX] = AE[X]$
- $E[b] = b$

Final Result

$$\mu_Y = A\mu_X + b$$

Reasoning

- Expectation distributes over addition and scalar/matrix multiplication.

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7. Correlation Matrix Transformation

7.1 Definition

$$R_{YY} = E[YY^T]$$

Substitute $Y = AX + b$:

$$R_{YY} = E[(AX + b)(AX + b)^T]$$

7.2 Expansion

$$R_{YY} = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

7.3 Apply Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$$

7.4 Final Result

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

Logical Flow

- Expand → apply expectation → substitute known terms.

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8. Covariance Matrix Transformation

8.1 Key Observation

$$Y - \mu_Y = (AX + b) - (A\mu_X + b)$$

$$Y - \mu_Y = A(X - \mu_X)$$

Reasoning

- The shift b cancels out.

8.2 Derivation

$$C_{YY} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

Substitute:

$$C_{YY} = E[A(X - \mu_X)(X - \mu_X)^T A^T]$$

8.3 Final Result

$$C_{YY} = AC_{XX}A^T$$

Important Conclusion

- **Shift b does not affect covariance.**

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9. Summary of Linear Transformation

If:

$$Y = AX + b$$

Then:

- Mean:

$$\mu_Y = A\mu_X + b$$

- Covariance:

$$C_{YY} = AC_{XX}A^T$$

- Correlation:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

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10. Numerical Example

Given

- $\mu_X = [2, -1, 1]^T$
- Matrix A
- Correlation matrix R_{XX}

10.1 Mean Vector

Using:

$$\mu_Y = A\mu_X$$

Result

$$\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

10.2 Correlation Matrix

Using:

$$R_{YY} = AR_{XX}A^T$$

Computation Steps

1. Compute AR_{XX}
2. Multiply by A^T

Result

$$R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

10.3 Covariance Matrix

Using:

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

Steps

1. Compute outer product $\mu_Y \mu_Y^T$
2. Subtract from R_{YY}

Result

$$C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

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11. Gaussian Distributions

11.1 Univariate Gaussian (1D)

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

11.2 Joint Gaussian (2D)

For two random variables:

- Defined using:
  - Means
  - Standard deviations
  - Correlation coefficient

11.3 Multivariate Gaussian (N-D)

For vector X:

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

Components

- Mean vector μ_X
- Covariance matrix C_{XX}

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12. Special Case: Uncorrelated Gaussian Variables

Condition

$$\mathrm{Cov}(X_i, X_j) = 0 \quad \forall i \neq j$$

Result

- Covariance matrix becomes **diagonal**.

$$C_{XX} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots \\ 0 & \sigma_2^2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Joint PDF

$$f_X(x) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

Interpretation

- Joint distribution factorizes into individual Gaussian PDFs.

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13. Homework

Problem

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$$

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14. Case Study: Smart City Traffic + Weather System

14.1 Step 1: Random Vector

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

Variables

- X_1: Traffic flow
- X_2: Speed
- X_3: Rainfall
- X_4: Temperature
- X_5: Pollution

14.2 Step 2: Mean Vector

$$\mu = E[X]$$

Represents **average system values**.

14.3 Step 3: Covariance Matrix

Represents **relationships between variables**.

Interpretation

- Rain ↑ → Traffic ↑
- Rain ↑ → Speed ↓
- Traffic ↑ → Pollution ↑
- Temperature ↑ → Pollution ↓

14.4 Step 4: Joint Gaussian Assumption

$$X \sim \mathcal{N}(\mu, \Sigma)$$

Meaning

System behavior is defined by:

- Mean → typical conditions
- Covariance → interdependencies

14.5 Step 6: Linear Transformation

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

Interpretation

- This is a **linear transformation**.

14.6 Inference

If:

$$X \sim \mathcal{N}(\mu, \Sigma)$$

Then:

- Y is also Gaussian
- Mean:

$$E[Y] = a^T \mu$$

- Variance:

$$\mathrm{Var}(Y) = a^T \Sigma a$$

14.7 Final Insight

If rainfall increases:

- Speed decreases
- Traffic increases
- Pollution increases

✅ COMPLETE LOGICAL FLOW OF LECTURE

1. Represent multiple variables as a **vector**
2. Define **mean, correlation, covariance**
3. Introduce **linear transformation** $Y = AX + b$
4. Derive how:
   - Mean transforms
   - Correlation transforms
   - Covariance transforms
5. Apply results to **numerical example**
6. Extend to **Gaussian distributions**
7. Generalize to **multivariate Gaussian**
8. Analyze **uncorrelated case**
9. Apply concepts to **real system (case study)**

🎯 FINAL EXAM TAKEAWAY

You must be able to:

- Define all matrices (mean, correlation, covariance)
- Derive transformation formulas

- Perform matrix computations
- Write Gaussian PDF forms
- Explain covariance interpretation
- Apply linear transformation results

If you want next:
✅ 1-page **ultra short revision sheet**
✅ **Important exam questions** from this lecture
✅ Step-by-step **proof of homework question**

Write a detailed lecture scribe for exam preparation based entirely on the given material. Include every concept, definition, and assumption, and clearly explain the reasoning behind each along with their logical connections. Ensure that all steps and arguments are presented in a structured and easy-to-follow manner. Do not add any external information, examples, or interpretations beyond the provided content. The final output should be clear, well-organized, and comprehensive, allowing a student to revise effectively and reconstruct all concepts and solutions without needing additional resources.

Here is your **fully detailed, structured lecture scribe** based **strictly and entirely on the provided PDF (Lecture 20)** .
All concepts, derivations, assumptions, and logical connections are included **without adding any external information**.

📘 Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

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1. Introduction and Scope

This lecture develops a structured framework for:

- Representing multiple random variables using vectors
- Defining statistical properties (mean, correlation, covariance)
- Studying how these properties change under linear transformations
- Extending to Gaussian distributions in multiple dimensions

The progression follows:

1. Representation
2. Statistical characterization
3. Transformation
4. Distribution modeling
5. Application

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2. Random Vector Representation

2.1 Definition

Given N random variables $X_1, X_2, \ldots, X_N$, define:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Key Points

- X is called a **random vector**
- It provides a **compact representation of multiple random variables**

Logical Role

- This representation enables applying **matrix operations** to multiple variables simultaneously

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3. Mean Vector

3.1 Definition

$$\mu_X = E[X]$$

3.2 Structure

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

3.3 Interpretation

- Represents the **average value of the random vector**
- Each element corresponds to the mean of a single variable

Logical Connection

- Extends scalar expectation to vector form **component-wise**

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4. Correlation Matrix

4.1 Definition

$$R_{XX} = E[XX^T]$$

4.2 Element-wise Form

$$R_{XX}(i,j) = E[X_i X_j]$$

4.3 Matrix Form

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots \\ E[X_2X_1] & E[X_2X_2] & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

4.4 Property

- R_{XX} is **symmetric**

Reasoning

- Since multiplication inside expectation is commutative:

$$E[X_iX_j] = E[X_jX_i]$$

---

5. Covariance Matrix

5.1 Definition

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

5.2 Expanded Form

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

5.3 Property

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

Logical Interpretation

- Covariance measures **relationships between variables**
- Diagonal → variances
- Off-diagonal → interdependencies

---

6. Linear Transformation of Random Vectors

6.1 Model

$$Y = AX + b$$

6.2 Components

- A: Transformation matrix
- b: Constant vector
- X: Input random vector

6.3 Interpretation

- A: scales and rotates data
- b: shifts/translates the center

6.4 Element-wise Expansion

$$Y_i = a_{i1}X_1 + a_{i2}X_2 + \cdots + a_{iN}X_N + b_i$$

Logical Meaning

- Each output variable depends linearly on all input variables

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7. Mean of Transformed Vector

7.1 Definition

$$\mu_Y = E[Y]$$

7.2 Derivation

Step 1: Substitute

$$\mu_Y = E[AX + b]$$

Step 2: Apply linearity

$$\mu_Y = E[AX] + E[b]$$

Step 3: Simplify

- $E[AX] = AE[X]$
- $E[b] = b$

7.3 Final Result

$$\mu_Y = A\mu_X + b$$

Logical Conclusion

- Mean transforms linearly
- Shift b directly affects the mean

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8. Correlation Matrix Transformation

8.1 Definition

$$R_{YY} = E[YY^T]$$

Substitute:

$$R_{YY} = E[(AX + b)(AX + b)^T]$$

8.2 Expansion

$$R_{YY} = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

8.3 Apply Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$$

8.4 Final Result

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

Logical Flow

- Expand → apply expectation → substitute definitions

---

9. Covariance Matrix Transformation

9.1 Key Observation

$$Y - \mu_Y = (AX + b) - (A\mu_X + b)$$

$$Y - \mu_Y = A(X - \mu_X)$$

Key Insight

- Shift b cancels out

9.2 Derivation

$$C_{YY} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

Substitute:

$$C_{YY} = E[A(X - \mu_X)(X - \mu_X)^T A^T]$$

9.3 Final Result

$$C_{YY} = AC_{XX}A^T$$

Important Conclusion

- Covariance depends only on A, not on b

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10. Summary of Transformation Results

If:

$$Y = AX + b$$

Then:

- Mean:

$$\mu_Y = A\mu_X + b$$

- Covariance:

$$C_{YY} = AC_{XX}A^T$$

- Correlation:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

---

11. Numerical Example

Given

- $\mu_X = [2, -1, 1]^T$
- Matrix A
- Matrix R_{XX}

11.1 Mean

$$\mu_Y = A\mu_X$$

Result

$$\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

11.2 Correlation

$$R_{YY} = AR_{XX}A^T$$

Steps

1. Compute AR_{XX}
2. Multiply by A^T

Result

$$R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

11.3 Covariance

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

Steps

1. Compute outer product
2. Subtract

Result

$$C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

---

12. Gaussian Distributions

12.1 Univariate Gaussian

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

12.2 Joint Gaussian (2D)

Defined using:

- Means
- Standard deviations
- Correlation coefficient

12.3 Multivariate Gaussian

For vector X:

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

Components

- Mean vector
- Covariance matrix

---

13. Special Case: Uncorrelated Gaussian Variables

13.1 Condition

$$\mathrm{Cov}(X_i, X_j) = 0 \quad \forall i \neq j$$

13.2 Result

- Covariance matrix becomes **diagonal**

13.3 Joint PDF

$$f_X(x) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n-\mu_n)^2}{2\sigma_n^2}\right)$$

Logical Meaning

- Joint distribution factors into individual distributions

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14. Homework

Statement

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$$

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15. Case Study: Smart City System

15.1 Random Vector

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

Variables

- Traffic flow
- Speed
- Rainfall
- Temperature
- Pollution

15.2 Mean Vector

$$\mu = E[X]$$

Represents average system values

15.3 Covariance Matrix

Represents relationships between variables

Observations

- Rain ↑ → Traffic ↑
- Rain ↑ → Speed ↓
- Traffic ↑ → Pollution ↑
- Temperature ↑ → Pollution ↓

15.4 Joint Gaussian Assumption

$$X \sim \mathcal{N}(\mu, \Sigma)$$

Meaning

System defined by:

- Mean (typical conditions)
- Covariance (interdependencies)

15.5 Linear Transformation

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

15.6 Result

If $X \sim \mathcal{N}(\mu, \Sigma)$:

- Y is Gaussian
- Mean:

$$E[Y] = a^T \mu$$

- Variance:

$$\mathrm{Var}(Y) = a^T \Sigma a$$

15.7 Final Insight

If rainfall increases:

- Speed decreases
- Traffic increases
- Pollution increases

✅ COMPLETE STRUCTURE OF LECTURE

1. Random vector representation
2. Mean vector
3. Correlation matrix

4. Covariance matrix
5. Linear transformation
6. Transformation of mean
7. Transformation of correlation
8. Transformation of covariance
9. Numerical example
10. Gaussian distributions
11. Special case (uncorrelated)
12. Case study application

🎯 FINAL EXAM PREPARATION NOTE

To reconstruct the entire lecture, you must be able to:

- Define all matrices
- Derive transformation results
- Perform matrix computations
- Write Gaussian PDFs
- Interpret covariance relationships
- Apply linear transformations

If you want, I can next give:

✅ **1-page quick revision sheet**
✅ **Most important exam questions**
✅ **Step-by-step derivations practice set**

Do you like this personality?